Basic commands

1. git init

Initializes a new Git repository in the current directory.

Example: git init

2. git clone

Copies a remote repository to the local machine.

Example: git clone https://github.com/user/repo.git

3. git status

Displays the state of the working directory and staging area.

Example: git status

4. git add

Adds changes to the staging area.

Example: git add file.txt

5. git commit

Records changes to the repository.

Example: git commit -m "Initial commit"

6. git config

Configures user settings, such as name and email.

Example: git config --global user.name "Your Name"

7. git log

Shows the commit history.

Example: git log

8. git show

Displays detailed information about a specific commit.

Example: git show <commit-hash>

9. git diff

Shows changes between commits, the working directory, and the staging area.

Example: git diff

10. git reset

Unstages changes or resets commits.

Example: git reset HEAD file.txt

Branching and Merging

11. git branch

Lists branches or creates a new branch.

Example: git branch feature-branch

12. git checkout

Switches between branches or restores files.

Example: git checkout feature-branch

13. git switch

Switches branches (modern alternative to git checkout).

Example: git switch feature-branch

14. git merge

Combines changes from one branch into another.

Example: git merge feature-branch

15. git rebase

Moves or combines commits from one branch onto another.

Example: git rebase main

16. git cherry-pick

Applies specific commits from one branch to another.

Example: git cherry-pick <commit-hash>

Remote Repositories

17. git remote

Manages remote repository connections.

Example: git remote add origin https://github.com/user/repo.git

18. git push

Sends changes to a remote repository.

Example: git push origin main

19. git pull

Fetches and merges changes from a remote repository.

Example: git pull origin main

20. git fetch

Downloads changes from a remote repository without merging.

Example: git fetch origin

21. git remote -v

Lists the URLs of remote repositories.

Example: git remote -v

Stashing and Cleaning

22. git stash

Temporarily saves changes not yet committed.

Example: git stash

23. git stash pop

Applies stashed changes and removes them from the stash list.

Example: git stash pop

24. git stash list

Lists all stashes.

Example: git stash list

25. git clean

Removes untracked files from the working directory.

Example: git clean -f

Tagging

26. git tag

Creates a tag for a specific commit.

Example: git tag -a v1.0 -m "Version 1.0"

27. git tag -d

Deletes a tag.

Example: git tag -d v1.0

28. git push --tags

Pushes tags to a remote repository.

Example: git push origin --tags

Advanced Commands

29. git bisect

Finds the commit that introduced a bug.

Example: git bisect start

30. git blame

Shows which commit and author modified each line of a file.

Example: git blame file.txt

31. git reflog

Shows a log of changes to the tip of branches.

Example: git reflog

32. git submodule

Manages external repositories as submodules.

Example: git submodule add https://github.com/user/repo.git

33. git archive

Creates an archive of the repository files.

Example: git archive --format=zip HEAD > archive.zip

34. git gc

Cleans up unnecessary files and optimizes the repository.

Example: git gc

GitHub-Specific Commands

35. gh auth login

Logs into GitHub via the command line.

Example: gh auth login

36. gh repo clone

Clones a GitHub repository.

Example: gh repo clone user/repo

37. gh issue list

Lists issues in a GitHub repository.

Example: gh issue list

38. gh pr create

Creates a pull request on GitHub.

Example: gh pr create --title "New Feature" --body "Description of the feature"

39. gh repo create

Creates a new GitHub repository.

Example: gh repo create my-repo